

3-2 (a) Formulate node-voltage equations for the circuit in Figure P3-2.

(b) Use these equations to find v_x and i_x .

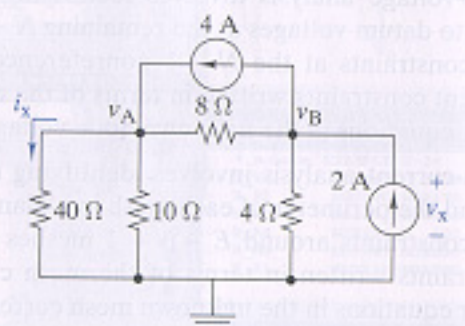


FIGURE P3-2

3-5 (a) Formulate node-voltage equations for the circuit in Figure P3-5.

(b) Solve for v_x and i_x when $R_1 = R_2 = R_3 = R_4 = 10 \text{ k}\Omega$, $v_s = 20 \text{ V}$, and $i_s = 2 \text{ mA}$.

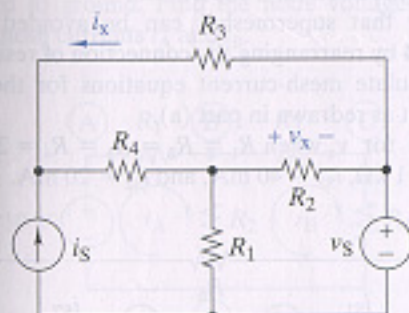


FIGURE P3-5

3-6 (a) Formulate node-voltage equations for the circuit in Figure P3-6.

(b) Solve for v_x and i_x when $R_1 = R_2 = R_3 = R_4 = R_5 = 10 \text{ k}\Omega$, and $v_s = 20 \text{ V}$.

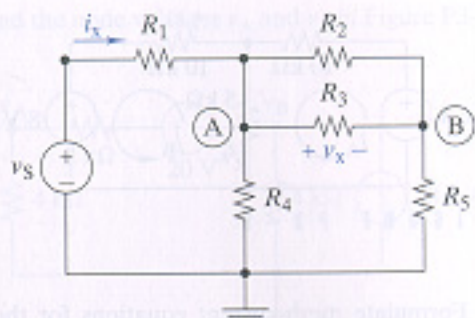


FIGURE P3-6

3-7 (a) Formulate node-voltage equations for the circuit in Figure P3-7.

(b) Solve for v_A , v_B , and v_C when $R_1 = 1 \text{ k}\Omega$, $R_2 = 2 \text{ k}\Omega$, $R_3 = 4 \text{ k}\Omega$, $R_4 = 2 \text{ k}\Omega$, and $i_{S1} = i_{S2} = 2 \text{ mA}$.

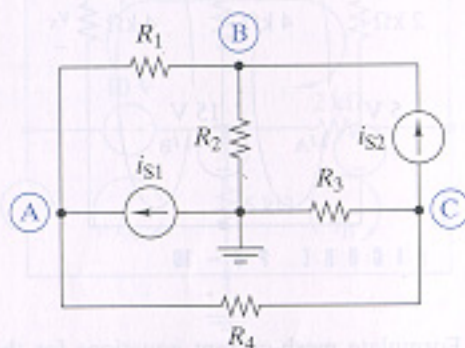


FIGURE P3-7

3-9 (a) Formulate mesh-current equations for the circuit in Figure P3-9.

(b) Use these equations to find v_x and i_x .

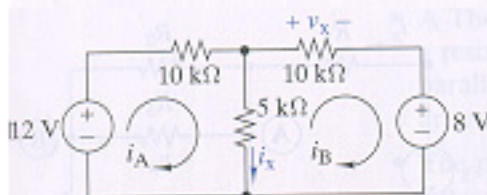


FIGURE P3-9

3-11 (a) Formulate mesh-current equations for the circuit in Figure P3-11.

(b) Use these equations to find v_x and i_x .

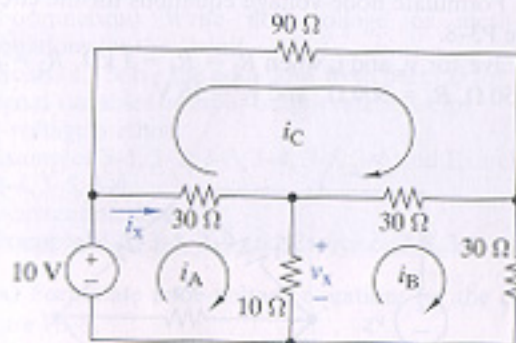


FIGURE P3-11

3-12 (a) Formulate mesh-current equations for the circuit in Figure P3-12.

(b) Solve for v_x and i_x when $R_1 = 200 \Omega$, $R_2 = 300 \Omega$, $R_3 = 50 \Omega$, $R_4 = 250 \Omega$, $R_5 = 200 \Omega$, $i_s = 50 \text{ mA}$, and $v_s = 15 \text{ V}$.

(c) Find the total power dissipated in the circuit.

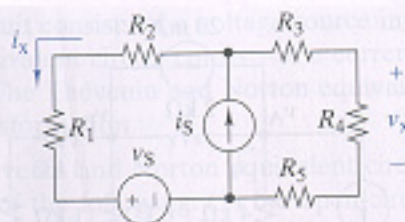


FIGURE P3-12

3-17 In Figure P3-17 all of the resistors are $1\text{ k}\Omega$. The voltage at node C is found to be $v_C = -2\text{ V}$ when node B is connected to ground. Find the node voltages v_A and v_D , and the mesh currents i_A and i_B .

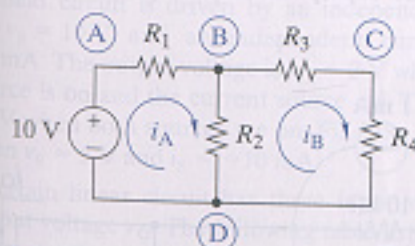


FIGURE P3-17

3-72 Attenuation Analysis

In Figure P3-72 a two-port attenuator connects a $600\text{-}\Omega$ source to a $600\text{-}\Omega$ load. Find the power delivered to the load in terms of v_S . Remove the attenuator and find the power delivered to the load when the source is directly connected to the load. By what fraction does the attenuator reduce the power delivered to the $600\text{-}\Omega$ load? Express the fraction in dB.

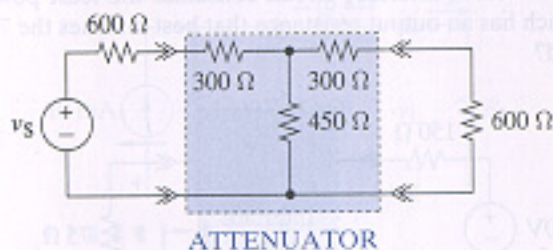


FIGURE P3-72

3-74 Battery Design

A satellite requires a battery with an open-circuit voltage $v_{OC} = 36\text{ V}$ and a Thévenin resistance $R_T \leq 10\text{ }\Omega$. The battery is to be constructed using series and parallel combinations of one of two types of cells. The first type has $v_{OC} = 9\text{ V}$, $R_T = 4\text{ }\Omega$, and a weight of 40 grams. The second type has $v_{OC} = 4\text{ V}$, $R_T = 0.5\text{ }\Omega$, and a weight of 15 grams. Design a minimum-weight battery that meets the open-circuit voltage and Thévenin resistance requirements.